Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	qwen3:8b

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Editorial Decision

MINOR REVISION

Weighted Score: 3.8 / 5.0
Confidence: 83%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	**5.0**	Excellent	■■■■■
Manuscript Structure	**2.5**	Weak	■■■■■
Methodology & Equations	**4.0**	Good	■■■■■
Statistical Analysis	**4.5**	Excellent	■■■■■
Results & Evidence	**4.2**	Good	■■■■■
Discussion & Conclusions	**3.5**	Good	■■■■■
Figures & Tables	**3.5**	Good	■■■■■
Scientific Writing	**3.5**	Good	■■■■■
References & Citations	**3.0**	Acceptable	■■■■■
Ethics & Transparency	**1.5**	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: **5.0 / 5.0** | Confidence: 95%

Strengths:

- [TITLE] clearly specifies the method and application domain
- [ABSTRACT] follows structured format with quantified results
- [KEYWORDS] include domain-specific terms and mix of general and specific terms

Minor Comments:

- [TITLE] could be slightly shortened for clarity without losing specificity

Recommendations:

- [TITLE] Consider shortening the title for improved readability while retaining technical specificity

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StructureReviewer

Manuscript structure and logical coherence

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical focus on physics-informed EANN framework
- Experimental validation of proposed framework
- Reproducibility emphasized through detailed methodology

Weaknesses:

- Missing literature review section
- Redundant content in section_vi and section_vii
- Research question/hypothesis not explicitly declared

Major Comments:

- The manuscript lacks a literature review section, which is critical for establishing the novelty and context of the work.
- Sections VI and VII are redundant and should be merged or removed.
- The research question or hypothesis is not explicitly declared in the introduction.

Minor Comments:

- Section flow could be improved with clearer transitions between sections.
- Some figures are referenced but not fully described in the text.

Recommendations:

- Add a comprehensive literature review section to contextualize the work.
- Merge or remove redundant sections VI and VII.
- Explicitly declare the research question or hypothesis in the introduction.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- Clear problem statement and motivation
- Robust experimental validation with statistical analysis
- Integration of physical constraints into the optimization framework

Weaknesses:

- The methodology section is not clearly structured to explain the EANN implementation and integration with physical constraints
- Equations are not consistently introduced or explained in context

Major Comments:

- The methodology section is not clearly structured to explain the EANN implementation and integration with physical constraints. This is critical for reproducibility and understanding the technical contribution.

Minor Comments:

- Equations are not consistently introduced or explained in context, which may hinder reproducibility.

Recommendations:

- Clarify the implementation of the EANN and how physical constraints are integrated into the neural network architecture.
- Ensure all equations are properly introduced and explained in context to improve reproducibility.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **4.5 / 5.0** | Confidence: 90%

Strengths:

- Use of Kruskal-Wallis and post-hoc Wilcoxon with Holm correction for non-normal data
- Bootstrap confidence intervals for robustness
- Coefficient of variation and RMSE/MSE for practical significance
- Pareto front and hypervolume indicators for multi-objective analysis

Weaknesses:

- No specific p-values reported for Kruskal-Wallis or post-hoc tests
- No mention of resample size for bootstrap confidence intervals

Major Comments:

- The manuscript should report p-values for Kruskal-Wallis and post-hoc Wilcoxon tests to fully support statistical inference.

Minor Comments:

- Consider specifying the number of resamples used in bootstrap confidence intervals for transparency.

Recommendations:

- Report p-values for all statistical tests to ensure clarity and reproducibility.
- Clarify the number of resamples used in bootstrap confidence intervals.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Equations are numbered sequentially and referenced in text
- Figures and tables are referenced in text before they appear
- The methodology is clearly described with mathematical formulations

Weaknesses:

- Some figures lack axis labels and units
- Some tables lack units and statistical significance indicators
- Some equations are not fully defined in the text

Major Comments:

- Equations (11), (13), and (16) are incomplete and need full definitions
- Figures 3, 4, and 5 lack axis labels and units for clarity
- Table X lacks units and statistical significance indicators

Minor Comments:

- Figure 6 does not indicate error bars or uncertainty in the Pareto front
- Equation (11) is not fully defined in the text

Recommendations:

- Add axis labels and units to Figures 3, 4, and 5
- Define equations (11), (13), and (16) in full
- Add units and statistical significance indicators to Table X

ResultsReviewer

Results validity and empirical support

Score: **4.2 / 5.0** | Confidence: 80%

Strengths:

- Clear presentation of MSE, RMSE, CV, and hypervolume metrics
- Explicit comparison with GA and PSO baselines
- Statistical validation with Kruskal-Wallis and Wilcoxon tests

Weaknesses:

- No explicit mention of negative results or limitations of the EANN framework
- Quantitative results are not consistently tied to figures or tables
- Lack of detailed reporting on the distribution of metrics across 30 runs

Major Comments:

- Negative results or limitations of the proposed method are not acknowledged, which is critical for reproducibility and scientific rigor.

Minor Comments:

- Quantitative results should be more clearly linked to figures and tables for clarity.

Recommendations:

- Include a section discussing limitations and potential failure cases of the EANN framework.
- Ensure all quantitative results (e.g., MSE, RMSE, CV) are explicitly tied to corresponding figures or tables.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear experimental validation and comparison with baselines (GA, PSO).
- Reproducibility is emphasized through statistical tests and multiple runs.

Weaknesses:

- Discussion lacks direct comparison with prior IEEE publications.
- Overgeneralization about practical utility for control deployment.

Major Comments:

- The discussion should explicitly compare the proposed method with prior IEEE publications to establish novelty.

Minor Comments:

- The conclusion could be more specific about the quantified improvements in performance.

Recommendations:

- Include a direct comparison with prior IEEE works in the discussion to highlight novelty.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical description of the EANN framework
- Well-defined experimental setup and validation methods
- Use of statistical validation techniques

Weaknesses:

- Overuse of passive voice
- Lack of definition for acronyms
- Vague quantifiers without numerical support

Major Comments:

- The manuscript needs to define acronyms such as EANN, PSO, and GA on first use.
- Passive voice is overused and should be minimized for clarity and directness.
- Quantifiers like 'significant' and 'many' should be replaced with numerical values for precision.

Minor Comments:

- Some sentences are run-on and could be split for better readability.
- The use of hedging language could be more consistent.

Recommendations:

- Define all acronyms on first use.
- Revise passive voice constructions to active voice where possible.
- Replace vague quantifiers with specific numerical values.
- Split long sentences for improved readability and clarity.

ReferencesReviewer

References quality, recency, and relevance

Score: **3.0 / 5.0** | Confidence: 80%

Strengths:

- Recent references from 2021-2023
- Diverse range of journals and conferences

Weaknesses:

- Lack of seminal foundational works in structural identification
- Missing key areas in literature review

Major Comments:

- The references section lacks seminal foundational works in structural identification, which is critical for establishing context and novelty.
- Missing key areas such as structural health monitoring and observer-based state reconstruction in the literature review.

Minor Comments:

- The book by Chopra is relevant but should be paired with more specific seminal works in the field of structural identification.

Recommendations:

- Include seminal papers on structural identification and observer design.
- Add references to key works in structural health monitoring and state estimation techniques.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.5 / 5.0** | Confidence: 90%

Strengths:

- The methodology and experimental results are well presented, with clear mathematical formulation and validation.

Weaknesses:

- Ethical considerations and limitations are not adequately addressed, which is a major concern for IEEE publication standards.

Major Comments:

- The manuscript lacks essential ethical reporting, including ethical approval, data availability, conflict of interest, and funding disclosure. These are required for IEEE publication and must be addressed before submission.

Minor Comments:

- The limitations section is missing, and reproducibility is not addressed, which affects the transparency and validity of the findings.

Recommendations:

- Authors must provide ethical approval details, data availability statement, conflict of interest declaration, and funding disclosure.
- Include a dedicated limitations section that explicitly and specifically addresses the study's constraints and their impact on the conclusions.
- Ensure reproducibility by providing access to data and code, as required by IEEE standards.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] The manuscript lacks a literature review section, which is critical for establishing the novelty and context of the work.

[2] Sections VI and VII are redundant and should be merged or removed.

[3] The research question or hypothesis is not explicitly declared in the introduction.

[4] The methodology section is not clearly structured to explain the EANN implementation and integration with physical constraints. This is critical for reproducibility and understanding the technical contribution.

[5] The manuscript should report p-values for Kruskal-Wallis and post-hoc Wilcoxon tests to fully support statistical inference.

[6] Equations (11), (13), and (16) are incomplete and need full definitions

[7] Figures 3, 4, and 5 lack axis labels and units for clarity

[8] Table X lacks units and statistical significance indicators

[9] Negative results or limitations of the proposed method are not acknowledged, which is critical for reproducibility and scientific rigor.

[10] The discussion should explicitly compare the proposed method with prior IEEE publications to establish novelty.

[11] The manuscript needs to define acronyms such as EANN, PSO, and GA on first use.

[12] Passive voice is overused and should be minimized for clarity and directness.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] [TITLE] could be slightly shortened for clarity without losing specificity

[2] Section flow could be improved with clearer transitions between sections.

[3] Some figures are referenced but not fully described in the text.

[4] Equations are not consistently introduced or explained in context, which may hinder reproducibility.

[5] Consider specifying the number of resamples used in bootstrap confidence intervals for transparency.

[6] Figure 6 does not indicate error bars or uncertainty in the Pareto front

[7] Equation (11) is not fully defined in the text

[8] Quantitative results should be more clearly linked to figures and tables for clarity.

[9] The conclusion could be more specific about the quantified improvements in performance.

[10] Some sentences are run-on and could be split for better readability.

[11] The use of hedging language could be more consistent.

[12] The book by Chopra is relevant but should be paired with more specific seminal works in the field of structural identification.

Specific Improvement Recommendations

1. [TITLE] Consider shortening the title for improved readability while retaining technical specificity
2. Add a comprehensive literature review section to contextualize the work.
3. Merge or remove redundant sections VI and VII.
4. Explicitly declare the research question or hypothesis in the introduction.
5. Clarify the implementation of the EANN and how physical constraints are integrated into the neural network architecture.
6. Ensure all equations are properly introduced and explained in context to improve reproducibility.

7. Report p-values for all statistical tests to ensure clarity and reproducibility.
8. Clarify the number of resamples used in bootstrap confidence intervals.
9. Add axis labels and units to Figures 3, 4, and 5
10. Define equations (11), (13), and (16) in full
11. Add units and statistical significance indicators to Table X
12. Include a section discussing limitations and potential failure cases of the EANN framework.